
a) Cereals
Yield
Profitability
Soil health
Inputs efficiency
Resilience to pests
Labour demand
Biodiversity
Climate mitigation
Biomass management (vs. no residues or mulch)
Organic mulch
69% n= 196
58% n= 12
76% n= 270
78% n= 54
69% n= 13
No data
0% n= 4
No data
Residues incorporation
67% n= 202
18% n= 11
67% n= 82
No data
No data
No data
No data
47% n= 30
Tree pruning applied
89% n= 38
40% n= 20
100% n= 6
No data
No data
No data
75% n= 4
No data
Residues retention
50% n= 4
50% n= 6
80% n= 10
No data
No data
No data
100% n= 1
0% n= 2
Nutrient management (vs. no fertilizer applied)
Organic fertilizer
66% n= 151
59% n= 27
90% n= 71
No data
No data
No data
75% n= 108
50% n= 4
Mixed fertilizer
84% n= 215
65% n= 20
70% n= 67
100% n= 4
No data
No data
No data
43% n= 23
Biochar amendment
63% n= 67
No data
0% n= 3
No data
No data
No data
No data
47% n= 15
Inoculation
60% n= 40
No data
No data
No data
42% n= 12
No data
No data
No data
Nutrient management (vs. inorganic fertilizer)
Mixed fertilizer
100% n= 17
No data
No data
No data
No data
No data
No data
No data
Organic fertilizer
100% n= 5
No data
No data
No data
No data
No data
No data
No data
Diversification (vs. monoculture)
Intercropping
76% n= 564
No data
No data
No data
96% n= 67
No data
71% n= 24
0% n= 3
Agroforestry
52% n= 175
50% n= 22
No data
No data
47% n= 32
No data
0% n= 5
No data
Crop rotation
49% n= 90
No data
No data
No data
No data
No data
50% n= 2
No data
Improved fallow
60% n= 40
No data
No data
No data
42% n= 12
No data
No data
No data
Natural Fallow
42% n= 40
No data
No data
No data
50% n= 12
No data
No data
No data
Water management (vs. no water harvesting)
Contour bunding
97% n= 36
67% n= 12
72% n= 498
79% n= 56
No data
No data
No data
No data
Microcatchments
100% n= 10
No data
75% n= 4
No data
No data
No data
No data
No data
Vegetative filter strips
100% n= 1
No data
No data
No data
No data
No data
No data
No data
Water management (vs. microcatchments)
No water harvesting
20% n= 10
No data
25% n= 4
No data
No data
No data
No data
No data
Soil management (vs. conventional tillage)
Reduced tillage
55% n= 82
73% n= 33
40% n= 15
No data
No data
100% n= 1
No data
No data
Variety management (vs. improved variety)
Local variety
25% n= 4
No data
No data
No data
No data
No data
No data
No data
Pest management (vs. mechanical method)
No Chemical Applied
No data
No data
No data
No data
No data
No data
0% n= 1
No data
b) Pulses and legumes
Diversification (vs. monoculture)
Intercropping
28% n= 254
69% n= 32
No data
No data
100% n= 11
No data
50% n= 20
0% n= 3
Agroforestry
27% n= 15
50% n= 22
No data
No data
No data
No data
No data
No data
Crop rotation
12% n= 8
No data
No data
No data
No data
No data
50% n= 2
No data
Biomass management (vs. no residues or mulch)
Residues incorporation
31% n= 152
18% n= 11
0% n= 18
No data
No data
No data
No data
No data
Residues retention
20% n= 5
No data
No data
No data
No data
No data
100% n= 3
No data
Organic mulch
0% n= 3
No data
No data
No data
No data
No data
No data
No data
Nutrient management (vs. no fertilizer applied)
Organic fertilizer
79% n= 66
82% n= 11
48% n= 64
No data
No data
No data
No data
No data
Biochar amendment
88% n= 24
No data
No data
No data
No data
No data
No data
No data
Mixed fertilizer
100% n= 24
No data
No data
No data
No data
No data
No data
No data
Variety management (vs. improved variety)
Local variety
32% n= 72
No data
50% n= 24
No data
No data
No data
No data
No data
Water management (vs. rainfed agriculture)
Supplemental irrigation
100% n= 12
No data
No data
11% n= 18
No data
No data
No data
No data
Pest management (vs. unspecified)
Synthetic pesticide
25% n= 8
75% n= 16
No data
No data
No data
No data
No data
No data
Soil management (vs. conventional tillage)
Zero tillage
0% n= 12
No data
No data
No data
No data
No data
No data
No data
Reduced tillage
0% n= 6
No data
No data
No data
No data
No data
No data
No data
c) Vegetables
Water management (vs. rainfed agriculture)
Supplemental irrigation
66% n= 402
50% n= 12
No data
No data
No data
No data
No data
No data
Diversification (vs. monoculture)
Intercropping
41% n= 22
No data
No data
No data
67% n= 12
No data
No data
No data
Nutrient management (vs. no fertilizer applied)
Mixed fertilizer
100% n= 2
75% n= 4
No data
No data
No data
No data
No data
0% n= 2
Organic fertilizer
100% n= 2
75% n= 4
No data
No data
No data
No data
No data
50% n= 2
d) Roots and tubers
Diversification (vs. monoculture)
Intercropping
12% n= 32
69% n= 32
No data
No data
100% n= 4
No data
50% n= 8
No data
Pest management (vs. unspecified)
Synthetic pesticide
50% n= 4
75% n= 16
No data
No data
No data
No data
No data
No data
Nutrient management (vs. no fertilizer applied)
Mixed fertilizer
0% n= 6
No data
No data
No data
No data
No data
No data
No data
e) Nuts and seeds
Diversification (vs. monoculture)
Intercropping
No data
No data
No data
No data
100% n= 6
No data
50% n= 12
No data
% positive comparisons
Negative (≤35%)
Neutral (36–64%)
Positive (≥65%)
Strong evidence (n ≥ 50, ≥ 3 studies)
Limited evidence (fewer comparisons or studies)